武汉大学 2012—2013 学年上学期

《C 语言程序设计》试卷（A）

学号：　　　　姓名：　　　　院系：　　　　专业：　　　　得分：

一、填空题（每空 2 分，总计 40 分）

（1）二进制数 110111.0101 转换成十进制数为：____________________，十进制数 424.75 转换成二进制数为：__________________，八进制数 41.27 转换成二进制数为：_____________________。

（2）下面程序中变量 a 的值为：__________，b 的值为：__________。

```
double x = 348.72,, y = -256.47;
int a = (int)(x + 0.50);
int b = (int)(y - 0.50);
```

（3）请写出下面程序段的输出结果：______________________________

。

```
#include <stdio.h>
const double PI = 3.14159265358979;
inline GetCircleArea(double radius){return PI*radius*radius;}

void main(){
    printf("Area = %.6lf\n", GetCircleArea(10.0));
}
```

（4）根据下面的代码，回答下面的问题

```
int a[] = {5, 15, 34, 54, 14, 2, 52, 72};
int *p = &a[1], *q = &a[5];
```

(p+3)的值是______；(q-3)的值是______；q-p 的值是______；p<q 的结果是真还是假______；*p < *q 的结果是真还是假______。

（5） 写出下面 main 函数运行结束时字符串 str 的内容：__________。

```
void main(){
    char str[] = “language”; int len = strlen(str);
    char *p1=str, *p2=str+len-1, t;
    while (p1 < p2){
        t=*p1; *p1=*p2; *p2=t;
        p1++; p2--;
    }
}
```

（6）程序中需要动态分配 1024 个 double 类型的数组，请在下面写出相应的内存分配语句和释放语句：

（7）在 C 语言中，&运算符作为单目运算符表示的是_________运算；作为双目运算符时表示的是_________运算。

（8）写出下面一段程序的运行结果：_________________________________。

```
inline void GetFilePath(char* strPath){
    int len = strlen(strPath);
    for (int i=len-1; i>=0; i--){
        if (strPath[i] == '\\'){
            strPath[i] = '\0';
            break;
        }
    }
}
void main(){
    char* strFile = "d:\\Data\\a.txt";
    GetFilePath(strFile);
    printf("%s\n", strFile);
}
```

（9）请写出下面程序段的输出结果：_________________________________

```
int i = 1;
switch (i % 3){
    case 0: printf("zero ");
    case 1: printf("one ");
    case 2: printf("two ");
}
```

（10）设有以下定义：

enum ColorList{NONE=-1,BLACK,RED,GREEN,BLUE,WHITE}color1;

则枚举类型的保留字是_________，用户定义的枚举类型名是_________，用户定义的枚举变量是_________。

二、改错题（共计 20 分）

图 1 所示为卫星影像的 RPC 参数文件（文本文件）。下面一段程序的功能是从硬盘文件上读取该参数文件并将文件内容显示在屏幕上。

```
LINE_OFF:     +8282.50000000000000000000 pixels
SAMP_OFF:     +8100.50000000000000000000 pixels
LAT_OFF:      +29.08086377429990000000 degrees
LONG_OFF:     +120.73213844246176000000 degrees
HEIGHT_OFF:   +4499.99544104747470000000 meters
LINE_SCALE:   +16204.00000000000000000000 pixels
SAMP_SCALE:   +16204.00000000000000000000 pixels
LAT_SCALE:    +5.63417751974040830000 degrees
LONG_SCALE:   +5.64465705982709660000 degrees
HEIGHT_SCALE: +9004.99099995009600000000 meters
LINE_NUM_COEFF_01: -2.299586477225247E-003
LINE_NUM_COEFF_02: -2.022684504636406E+000
LINE_NUM_COEFF_03: -1.074561934042836E+001
LINE_NUM_COEFF_04: -7.077590911389452E-002
LINE_NUM_COEFF_05: +1.067012394973569E-002
LINE_NUM_COEFF_06: -5.407548570510540E-005
LINE_NUM_COEFF_07: +1.334418341743077E-007
LINE_NUM_COEFF_08: -5.065827199711402E-002
LINE_NUM_COEFF_09: +9.787596307938647E-004
LINE_NUM_COEFF_10: +1.106914577378022E-004
LINE_NUM_COEFF_11: -7.274127052060035E-006
LINE_NUM_COEFF_12: +2.612788902560689E-004
LINE_NUM_COEFF_13: -5.595537071517691E-006
LINE_NUM_COEFF_14: +1.425244022753086E-004
LINE_NUM_COEFF_15: -8.705078456582647E-004
LINE_NUM_COEFF_16: -1.846082502160901E-003
LINE_NUM_COEFF_17: +7.559431121245367E-004
LINE_NUM_COEFF_18: +6.289113528395995E-006
LINE_NUM_COEFF_19: -3.231376365409801E-005
LINE_NUM_COEFF_20: +4.801059434609350E-006
LINE_DEN_COEFF_01: +1.000000000000000E+000
LINE_DEN_COEFF_02: +7.761539939371039E-003
LINE_DEN_COEFF_03: +7.436283074271652E-003
LINE_DEN_COEFF_04: +4.361237683715317E-005
LINE_DEN_COEFF_05: -1.795163419341408E-003
LINE_DEN_COEFF_06: -4.077671874606578E-005
LINE_DEN_COEFF_07: -2.561073600148974E-004
LINE_DEN_COEFF_08: -9.003460548404828E-003
LINE_DEN_COEFF_09: -1.989905014332467E-002
LINE_DEN_COEFF_10: -7.145879337097903E-005
LINE_DEN_COEFF_11: +9.948547804417857E-005
LINE_DEN_COEFF_12: -3.559736601822449E-004
LINE_DEN_COEFF_13: -1.511892844527223E-004
LINE_DEN_COEFF_14: -1.338653514548384E-006
LINE_DEN_COEFF_15: -4.228821559922279E-004
LINE_DEN_COEFF_16: -8.662370789721402E-004
LINE_DEN_COEFF_17: -1.398766257312492E-006
LINE_DEN_COEFF_18: -4.925486605563726E-004
LINE_DEN_COEFF_19: -1.269276958523937E-004
LINE_DEN_COEFF_20: -4.226001610342639E-009
SAMP_NUM_COEFF_01: +5.911186628436804E-003
SAMP_NUM_COEFF_02: +1.045790856554556E+001
SAMP_NUM_COEFF_03: -2.483367845676971E+000
SAMP_NUM_COEFF_04: +5.667872833776894E-004
SAMP_NUM_COEFF_05: -2.469575721630076E-002
SAMP_NUM_COEFF_06: +1.780789141996727E-003
SAMP_NUM_COEFF_07: -3.233957829356492E-004
SAMP_NUM_COEFF_08: -1.226407241970775E-002
SAMP_NUM_COEFF_09: -8.872948052688130E-002
SAMP_NUM_COEFF_10: -2.517510603930681E-006
SAMP_NUM_COEFF_11: +1.861730598589650E-004
SAMP_NUM_COEFF_12: -2.173831003818562E-003
SAMP_NUM_COEFF_13: +1.168918282006353E-003
SAMP_NUM_COEFF_14: -9.767548212679567E-004
SAMP_NUM_COEFF_15: +9.224495836824383E-004
SAMP_NUM_COEFF_16: +6.306763587629046E-003
SAMP_NUM_COEFF_17: +2.341159827299103E-004
SAMP_NUM_COEFF_18: +4.878990604180726E-005
SAMP_NUM_COEFF_19: +3.281542592259337E-004
SAMP_NUM_COEFF_20: -4.338841047605151E-008
SAMP_DEN_COEFF_01: +1.000000000000000E+000
SAMP_DEN_COEFF_02: +5.624289135473920E-003
SAMP_DEN_COEFF_03: +5.344824541257472E-002
SAMP_DEN_COEFF_04: -1.950128568352968E-002
SAMP_DEN_COEFF_05: -6.642535299175518E-003
SAMP_DEN_COEFF_06: -4.514648645300367E-005
SAMP_DEN_COEFF_07: -1.053324506254098E-003
SAMP_DEN_COEFF_08: +2.498255453198464E-002
SAMP_DEN_COEFF_09: +3.846926022441363E-003
SAMP_DEN_COEFF_10: -6.682497431532682E-005
SAMP_DEN_COEFF_11: +1.687179241756267E-004
SAMP_DEN_COEFF_12: -2.986554860439318E-005
SAMP_DEN_COEFF_13: +5.830216576292940E-004
SAMP_DEN_COEFF_14: -6.004704106351078E-007
SAMP_DEN_COEFF_15: +3.740970997376860E-004
SAMP_DEN_COEFF_16: -3.551902358761580E-004
SAMP_DEN_COEFF_17: +2.293785351289550E-006
SAMP_DEN_COEFF_18: -3.761939610564053E-004
SAMP_DEN_COEFF_19: +2.990218033208137E-004
SAMP_DEN_COEFF_20: +1.826691236339099E-006
```

图 1 卫星影像的 RPC 参数文件样例

```c
#include <stdio.h>

typedef struct tagRPCPARA{
    double line_off,  line_sca;
    double samp_off,  samp_sca;
    double lon_off,   lon_sca;
    double lat_off,   lat_sca;
    double hig_off,   hig_sca;
    double line_num_coef[20], line_den_coef[20];
    double samp_num_coef[20], samp_den_coef[20];
}RpcPara;

bool ReadRpcFile(const char* strFile, RpcPara* pRPC){
    FILE* fp = fopen(strFile, "rt");
    if (!fp){
        printf("Can not open file %s\n", strFile);                错误(1)
        return;
    }

    char str[32];
    fscanf(fp, "%s %lf %s", str, &pRPC->line_off, str);
    fscanf(fp, "%s %lf %s", str, &pRPC->samp_off, str);
    fscanf(fp, "%s %lf %s", str, &pRPC->lat_off,  str);
    fscanf(fp, "%s %lf %s", str, &pRPC->lon_off,  str);
    fscanf(fp, "%s %lf %s", str, &pRPC->hig_off,  str);
    fscanf(fp, "%s %lf %s", str, &pRPC->line_sca, str);
    fscanf(fp, "%s %lf %s", str, &pRPC->samp_sca, str);
    fscanf(fp, "%s %lf %s", str, &pRPC->lat_sca,  str);
    fscanf(fp, "%s %lf %s", str, &pRPC->lon_sca,  str);
    fscanf(fp, "%s %lf %s", str, &pRPC->hig_sca,  str);

    for (i=0; i<20; i++)                                          错误(2)
        fscanf(fp, "%s %lf", str, &pRPC->line_num_coef[i])
    for (i=0; i<20; i++)                                          错误(3)
        fscanf(fp, "%s %lf", str, &pRPC->line_den_coef[i]);
    for (i=0; i<20; i++)
        fscanf(fp, "%s %lf", str, &pRPC->samp_num_coef[i]);
    for (i=0; i<20; i++)
        fscanf(fp, "%s %lf", str, &pRPC->samp_den_coef[i]);

    return true;
}
```

```
void main(){
    RpcPara rpc;
    if (ReadRpcFile("D:\ZY3-MATCH\73_BWD.rpc", &Rpc){    警告(4)；错误(5, 6)
        printf("LINE_OFF:       %.20lf pixels\n", rpc.line_off);
        printf("SAMP_OFF:       %.20lf pixels\n", rpc.samp_off);
        printf("LAT_OFF:        %.20lf pixels\n", rpc.lat_off);
        printf("LONG_OFF:       %.20lf pixels\n", rpc.lon_off);
        printf("HEIGHT_OFF:     %.20lf pixels\n", rpc.hig_off);
        printf("LINE_SCALE:     %.20lf pixels\n", rpc.Line_sca);
        printf("SAMP_SCALE:     %.20lf pixels\n", rpc.samp_sca);
        printf("LAT_SCALE:      %.20lf pixels\n", rpc.lat_sca);
        printf("LONG_SCALE:     %.20lf pixels\n", rpc.lon_sca);
        printf("HEIGHT_SCALE:   %.20lf pixels\n", rpc.hig_sca);

        for (int i=0; i<20; i++)
            printf("LINE_NUM_COEFF_%d: %.15e\n", i+1, rpc.line_num_coef[i]);
        for (int i=0; i<20; i++)
            printf("LINE_DEN_COEFF_%d: %.15e\n", i+1, rpc.line_den_coef[i]);
        for (int i=0; i<20; i++)
            printf("SAMP_NUM_COEFF_%d: %.15e\n", i+1, rpc.samp_num_coef[i]);
        for (int i=0; i<20; i++)
            printf("SAMP_DEN_COEFF_%d: %.15e\n", i+1, rpc.samp_den_coef[i]);
    }
}
```

1、请根据下述错误提示信息修改程序中的编译错误（总计 12 分）;

(1) error C2065: 'ReadRpcFile' : function must return a value
(2) error C2065: 'i' : undeclared identifier
(3) error C2143: syntax error : missing ';' before 'for'
(4) warning C4129: 'Z' : unrecognized character escape sequence
(5) error C2065: 'Rpc' : undeclared identifier
(6) error C2143: syntax error : missing ')' before '{'
(7) error C2039: 'Line_scale' : is not a member of 'tagRPCPARA'

2、程序中另有两处逻辑错误，请查找并修改。（8 分）

三、编程题（每题 10 分，共计 40 分）

1、编写程序，求 1 - 3 + 5 - 7 + 9 - 11 + ... - 99 + 101 的值；

2、输入 n 个评委的评分，计算并输出参赛选手的最后得分。计算方法为去除一个最高分和最低分，其余的进行平均，得出参赛选手的最后得分。

3、编写一个递归函数创建多级目录的文件夹。其中函数的声明方式如下：

```
void CreateDirs(const char *strPath);
```

函数的调用方式如下：

```
#include <stdio.h>
#include <direct.h>
#include <string.h>

void main(){
    CreateDirs("D:\\Level0\\Level1\\Level2\\Level3");
}
```

其中需要使用库函数 _mkdir，其在 MSDN 中的帮助信息如下：

Creates a new directory.

int _mkdir(const char *dirname);

Parameters

dirname: Path for a new directory.

Return Value

Each of these functions returns the value 0 if the new directory was created. On an error, the function returns –1 and sets errno as follows.

#include <direct.h>

4、如图 2 所示是一张成绩单，其中第一列是学生姓名，第二列是学生的学号，第三列是数学成绩。请根据题目后面的程序片段补出相应的三个函数代码。

张三	2590028	82
李四	2590030	80
王五	2590032	75
刘六	2590033	76
田七	2590036	91
...		

图 2 成绩单示意图

```
#include <stdio.h>
#include <string.h>

struct STU;
typedef STU* STUPTR;

struct STU{
    int ID;
    char name[16];
    float score;
    STUPTR pNext;
};

// 读取成绩单文件，获得学生的成绩
bool ReadScoresFile(const char* strFile, STUPTR* pSTU);

// 在屏幕上输出学生的成绩单
void PrintScores(STUPTR pSTU);

// 释放成绩单的数据结构
void ReleaseScores(STUPTR pSTU);

void main(){
    STUPTR pSTU = NULL;
    if (ReadScoresFile("d:\\scores.txt", &pSTU)){
        PrintScores(pSTU);
        ReleaseScores(pSTU);
    }
}
```

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